

COSC 594

Unlearning: DM

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Announcement

Check Canvas for Homework 3

Due on 10/24, 11:59pm

Team assignment

One volunteer is needed to work independently

Outline

- MOTIVATION
- EXAMPLES
- REGULARIZATION-BASED APPROACH
- CONSTRAINED APPROACH

Outline

■ MOTIVATION

The Right to be Forgotten

The General Data Protection Regulation (GDPR)

Recitals 65 and 66 and in Article 17

“The data subject shall have the right to obtain from the controller
the erasure of personal data concerning him or her without undue delay and
the controller shall have the obligation to erase personal data without undue delay”

Increasing concerns on copyright

■ STABILITY AI



HOME > BUSINESS > BUSINESS NEWS

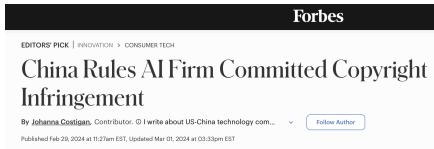
Artists Score Major Win in Copyright Case Against AI Art Generators

The court declined to dismiss copyright infringement claims against the AI companies. The order could implicate other firms that used Stable Diffusion, the AI model at issue in the case.

BY WINSTON CHO  AUGUST 13, 2024 1:09PM

■ CHINESE FIRMS

CHINA **IP Law Update**
中国知识产权法律更新



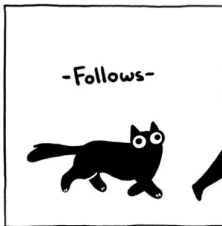
Chinese Court Again Rules AI-Generated Images Are Eligible for Copyright Protection

Post Date:
03/14/2025

Author:
Aaron Winger

On March 7, 2025, the Changshu People's Court announced that it had ruled that images generated with Artificial Intelligence (AI) are eligible for copyright protection. This is believed to be the second case regarding AI-generated images with the **Beijing Internet Court** ruled similarly in late 2023. In the instant case, Lin XX generated an

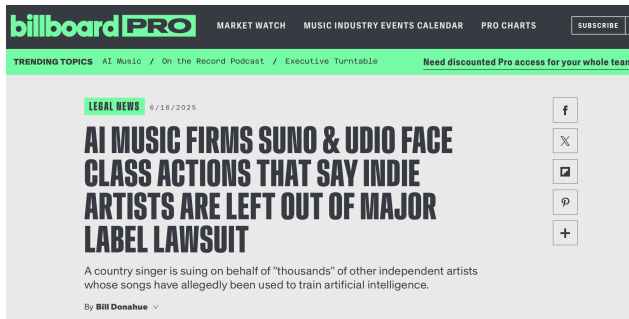
A case of Andersen's comics and Stability AI's recreation from late 2022



© Sarah Andersen

Stability AI vs Sarah Andersen

Developing cases of singers and AI music firms



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Oct 7, 2025 • 3 min read ★

Anthropic's own guardrails revive full copyright lawsuit from litigious music publishers

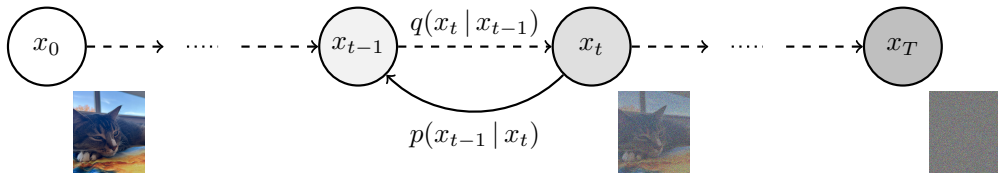
Earlier this year a judge dismissed all but one of the copyright claims in a lawsuit filed by three music publishers against AI company Anthropic. However, the publishers revised and resubmitted those claims, and this time the judge reckons all the allegations are sufficiently "plausible" to proceed

Outline

■ MOTIVATION

■ EXAMPLES

Diffusion model



data point x_0 – image, trajectory, video, protein

■ ANALOGUE W/ VAE

- ★ input and latent

$$x = x_0 \quad \text{and} \quad z_{1:T} = x_{1:T}$$

- ★ noiser (encoder)

$$q(x_{1:T} | x_0) = q(x_1 | x_0)q(x_2 | x_1) \dots q(x_T | x_{T-1})$$

- ★ denoiser (decoder)

$$p(x_0 | x_{1:T}) = p(x_T)p(x_{T-1} | x_T) \dots p(x_0 | x_1)$$

■ GROUND-TRUTH DENOISER

$$q(x_{t-1} | x_t, x_0) = \mathcal{N}(x_{t-1}; \mu_q(x_t, x_0), \sigma^2(t)I)$$

$$q(x_t | x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1 - \alpha_t)I)$$

$$\mu_q(x_t, x_0) = \frac{1}{\sqrt{\alpha_t}}x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\alpha_t}}\epsilon_0 - \text{noise-based mean}$$

$$\sigma^2(t) - \text{variance}$$

■ APPROXIMATE DENOISER

$$p_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma^2(t)I), \quad \mu_\theta(x_t, t) = \frac{1}{\sqrt{\alpha_t}}x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}\sqrt{\alpha_t}}\epsilon_\theta(x_t, t)$$

■ MEAN SQUARE MINIMIZATION

$$\underset{\theta}{\text{minimize}} \quad D_{\text{KL}}(q(x_{t-1} | x_t, x_0) \| p_\theta(x_{t-1} | x_t)) \quad \Leftrightarrow \quad \underset{\theta}{\text{minimize}} \quad \|\epsilon_0 - \epsilon_\theta(x_t, t)\|^2$$

$$\text{Pretrained model } p_t^{\text{pre}}(x_{t-1} | x_t) = p_{\theta^*}(x_{t-1} | x_t)$$

Forget quality and model utility for unlearning

Conditional diffusion model: $\epsilon_{\theta}(x_t, t; c)$ w/ parameter θ , condition c

Forget data set: $\mathcal{D}^f = \{(x_0^i, c^i)\}_{i=1}^m$

Retain data set: $\mathcal{D}^r = \{(x_0^j, c^j)\}_{j=1}^n$

■ FORGET QUALITY

“unlearn the forget data set”

■ MODEL UTILITY

“maintain the model’s performance on the retain data set”

No consensus on right metrics

Tradeoff between forget quality and model utility

Retaining objective

Conditional diffusion model: $\epsilon_{\theta}(x_t, t; c)$ w/ parameter θ , condition c

Forget data set: $\mathcal{D}^f = \{(x_0^i, c^i)\}_{i=1}^m$

Retain data set: $\mathcal{D}^r = \{(x_0^j, c^j)\}_{j=1}^n$

■ RETAINING OBJECTIVE

$$\underset{\theta}{\text{minimize}} \quad \mathbb{E}_{(x_0, c) \sim \mathcal{D}^r} [\|\epsilon_0 - \epsilon_{\theta}(x_t, t; c)\|^2]$$

Forgetting objectives

Conditional diffusion model: $\epsilon_{\theta}(x_t, t; c)$ w/ parameter θ , condition c

Forget data set: $\mathcal{D}^f = \{(x_0^i, c^i)\}_{i=1}^m$

Retain data set: $\mathcal{D}^r = \{(x_0^j, c^j)\}_{j=1}^n$

■ FORGETTING OBJECTIVE #1

$$\underset{\theta}{\text{maximize}} \quad \mathbb{E}_{(x_0, c) \sim \mathcal{D}^f} [\|\epsilon_0 - \epsilon_{\theta}(x_t, t; c)\|^2]$$

■ FORGETTING OBJECTIVE #2

$$\underset{\theta}{\text{minimize}} \quad \mathbb{E}_{(x_0, c) \sim \mathcal{D}^f} [\|\epsilon_0^f - \epsilon_{\theta}(x_t, t; c)\|^2]$$

$$\epsilon_0^f = \epsilon_{\theta}(x_t, t; \tilde{c}) \text{ for } \tilde{c} \notin \mathcal{D}^f$$

Gradient-based update

$$\theta^+ \leftarrow \theta + \eta \nabla_{\theta} \mathbb{E}_{(x_0, c) \sim \mathcal{D}^t} [\|\epsilon_0 - \epsilon_{\theta}(x_t, t; c)\|^2]$$

η – stepsize

Unlearning $\equiv$ **reverse** of model training/learning

Forget nudity or class

SD v1.4



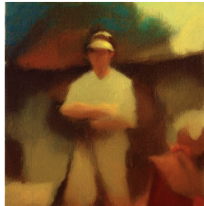
ESD



SA



SalUn



EraseDiff



★ Added by authors for publication

Tradeoff between forget quality and model utility

example source: [arXiv:2401.05779](https://arxiv.org/abs/2401.05779)

Outline

- MOTIVATION
- EXAMPLES
- REGULARIZATION-BASED APPROACH

Forgetting objective

Diffusion model: $x_0 \sim p(\cdot)$

Forget distribution: $x_0^i \sim p^i(\cdot)$ for $i = 1, \dots, m$

Retain distribution: $x_0^j \sim p^j(\cdot)$ for $j = 1, \dots, n$

■ FORGETTING OBJECTIVE

$$\underset{p}{\text{minimize}} \quad \mathbb{E}_{x_0 \sim p^i(\cdot)} [\log p(x_0)]$$



$$\underset{p}{\text{maximize}} \quad D_{\text{KL}}(p^i(x_0) \parallel p(x_0))$$

Retaining objective

Diffusion model: $x_0 \sim p(\cdot)$

Forget distribution: $x_0^i \sim p^i(\cdot)$ for $i = 1, \dots, m$

Retain distribution: $x_0^j \sim p^j(\cdot)$ for $j = 1, \dots, n$

■ RETAINING OBJECTIVE

$$\underset{p}{\text{maximize}} \quad \mathbb{E}_{x_0 \sim p^j(\cdot)} [\log p(x_0)]$$



$$\underset{p}{\text{minimize}} \quad D_{\text{KL}}(p^j(x_0) \parallel p(x_0))$$

Joint optimization

$$\underset{p}{\text{minimize}} \quad -D_{\text{KL}}(p^i(x_0) \parallel p(x_0)) + D_{\text{KL}}(p^j(x_0) \parallel p(x_0))$$

Leave away from forget distribution and stay close to retain distribution

■ DECOUPLED CASE

$$\text{supp}(p^i) \cap \text{supp}(p^j) = \emptyset$$

$\Downarrow$

$$p^*(\cdot) = \begin{cases} 0 & \text{when } \exists x_0 \in \text{supp}(p^i) \\ p^j(\cdot) & \text{when } x_0 \in \text{supp}(p^j) \end{cases}$$

Unknown forget quality

Relaxation to concept unlearning

Diffusion model: $x_0 \sim p(\cdot)$

Forget distribution: $x_0^f \sim p^f(\cdot)$

Retain distribution: $x_0^r \sim p^r(\cdot)$

p^f, p^r – e.g., finetuned models

■ FORGETTING OBJECTIVE

$$\underset{p}{\text{maximize}} \quad D_{\text{KL}}(p(\cdot) \parallel p^f(\cdot))$$

■ RETAINING OBJECTIVE

$$\underset{p}{\text{minimize}} \quad D_{\text{KL}}(p(\cdot) \parallel p^r(\cdot))$$

Joint policy optimization

$$\underset{p}{\text{minimize}} \quad -D_{\text{KL}}(p(\cdot) \parallel p^{\text{f}}(\cdot)) + D_{\text{KL}}(p(\cdot) \parallel p^{\text{r}}(\cdot))$$

Leave away from forget distribution and stay close to retain distribution

■ OPTIMAL MODEL

$$p^{\star}(\cdot) = \frac{p^{\text{r}}(\cdot)}{p^{\text{f}}(\cdot)} \frac{1}{Z}$$

Negation! $\implies$ Determined forget quality!

Outline

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Constrained policy optimization

$$\begin{array}{ll} \underset{p}{\text{minimize}} & D_{\text{KL}}(p(\cdot) \parallel p^{\text{r}}(\cdot)) \\ \text{subject to} & D_{\text{KL}}(p(\cdot) \parallel p^{\text{f}}(\cdot)) \geq \epsilon \end{array}$$

Stay close to retain distribution and leave away from forget distribution by ϵ

■ OPTIMAL MODEL

$$p^{\star}(\cdot) = \frac{(p^{\text{r}}(\cdot))^{\frac{1}{1-\lambda^{\star}}}}{(p^{\text{f}}(\cdot))^{\frac{\lambda^{\star}}{1-\lambda^{\star}}}} \frac{1}{Z}$$

When strong duality holds and an optimal dual variable is $\lambda^{\star} < 1$

Negation! $\implies$ Determined forget quality!

Summary

■ MOTIVATION

Right to be forgotten, copyright concerns, developing lawsuits

■ EXAMPLES

Reverse of model training/learning

■ REGULARIZATION-BASED APPROACH

Forward KL leads to unknown forget quality, reverse KL yields determined forget quality

■ CONSTRAINED APPROACH

Nonconvex problem